PHYSICAL CUMULATIVE EXAMINATION

April 28, 1990

One of the less common radioactive decay modes is electron capture, in which a proton in the atomic nucleus reacts with one of the electrons of the atom. In the process, the atomic number of the element decreases by one, while the atomic mass remains approximately constant. An important example is the decay of cobalt-56 to give iron-56 and gamma rays.

The reaction takes place only if an electron is within a nucleus, as the nuclear force responsible for the reaction has a very short range. This means that practically only the electrons in the K shell (the 1s electrons) react.

Estimate the probability that a 1s electron in the cobalt atom will be within the nucleus. The radius of the Co nucleus is approximately 4.5×10^{-15} m.

The wave function for the 1s orbital is, in good approximation,

$$\psi(r) = \pi^{-1/2}(Z/a_0)^{3/2}e^{-\frac{Z}{a_0}r}$$

where $a_0 \simeq 0.529\times10^{-10}$ m is the atomic unit of length, and Z is the atomic number. Z = 27 for cobalt.

The approximate formula

$$\int_0^R r^n e^{-cr}dr \approx \frac{R^{n+1}}{n+1} - c\frac{R^{n+2}}{n+2}$$

may be useful. This formula is valid if $R \ll \frac{1}{c}$.

Do not forget that the volume element for spherical integration is

$$dV = 4\pi r^2 dr.$$